

OnTime+ — A Schedule-Aware Transit Assistant

1. Motivation

Public transportation systems like MBTA are often unpredictable due to delays, service changes, and disruptions. For UMass Boston students who rely on public transit, arriving on time for classes, exams, or meetings can be challenging.

Most existing tools (e.g., Google Maps) focus on route planning but do not consider user schedules or provide reliability-aware recommendations under uncertainty.

2. Target User

A UMass Boston student who depends on MBTA for daily commuting and needs to arrive on time reliably, especially when facing possible service disruptions.

3. Problem Statement

Will I arrive on time, and what is the safest way to ensure that?

This transforms the task into a time-constrained decision-making problem under uncertainty.

4. Proposed System

We propose a RAG-based chatbot that integrates user schedules (e.g., class time) and transit information (routes, delays, alternatives).

The system will recommend when to leave, suggest the most reliable route, and provide risk-aware guidance.

5. Core Features

Schedule-Aware Planning: Input event time and output recommended departure time with buffer.

On-Time Estimation: Answer queries such as 'If I leave now, will I be on time?'

Disruption-Aware Routing: Incorporate transit alerts and suggest alternatives.

Risk-Aware Recommendations: Label outputs as reliable or risky.

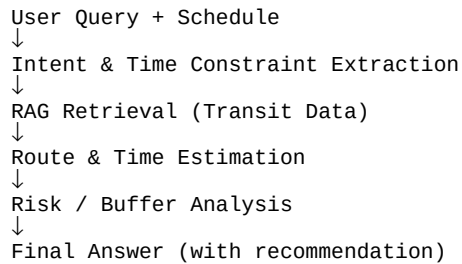
6. Data Sources (RAG)

Real / Downloadable Sources: MBTA alerts, route and station information, UMass Boston shuttle information.

Constructed / Simulated Data: Typical travel times, predefined disruption scenarios, and user schedule input.

This approach allows realistic simulation without external APIs.

7. System Architecture (High-Level)



8. Example Queries

My class starts at 10 AM—when should I leave?

If I leave at 9:30, will I be on time?

The Red Line is delayed—what should I do?

What is the safest way to ensure I arrive on time?

9. Expected Contribution

The system integrates user schedules, supports decision-making under uncertainty, and provides reliability-aware recommendations beyond traditional route planning.

10. Next Steps

Finalize user definition and scope, collect or build the RAG dataset, and design the pipeline and prototype.